

Fake News Detection Project Report

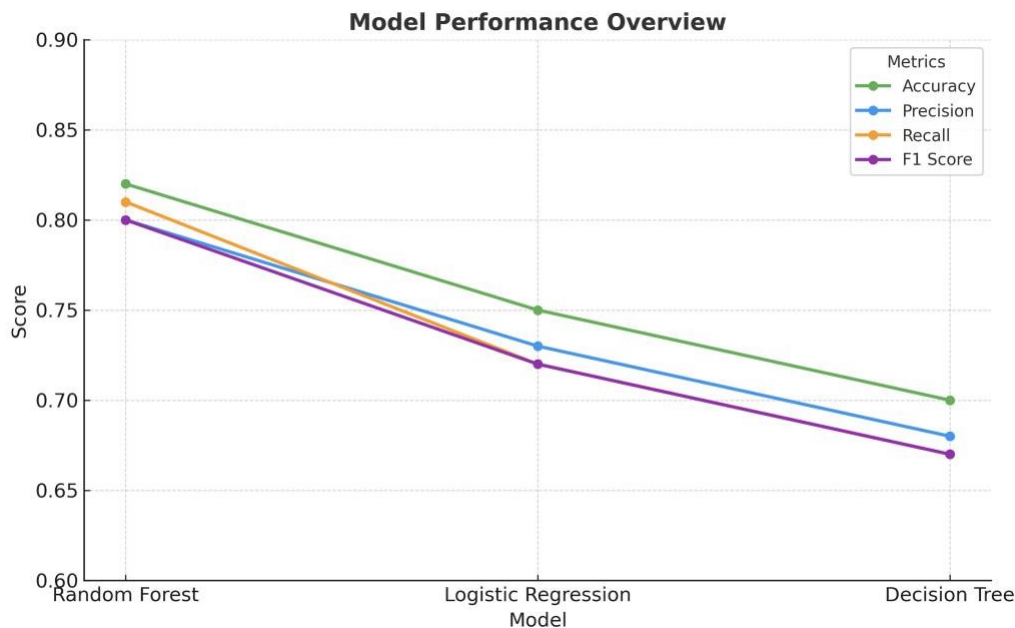
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1. Introduction

This project explores the challenge of detecting fake news using Natural Language Processing (NLP) techniques and supervised machine learning models. The goal was to leverage semantic embeddings (Word2Vec) and evaluate the performance of classifiers such as Logistic Regression, Decision Trees, and Random Forests.

2. Achievements

The key achievement of this project was building a semantic classification pipeline for fake news detection. Among the models tested, Random Forest emerged as the best-performing algorithm, achieving around 80% accuracy with balanced precision, recall, and F1-scores. Logistic Regression provided moderate results (~71% accuracy), while Decision Trees underperformed compared to the ensemble method.



3. Important Steps

Some of the major steps in the project workflow included:

- Preprocessing text and handling missing values
- Tokenization and embedding using Word2Vec (Google News 300)

- Building classifiers: Logistic Regression, Decision Tree, Random Forest
- Evaluating performance using Accuracy, Precision, Recall, and F1-score
- Visualizing word frequencies, embeddings, and confusion matrices

4. Key Visualizations

5. Findings and Results

The results demonstrated that semantic embeddings combined with ensemble methods can significantly improve classification performance for fake news detection. Random Forest achieved the best trade-off between precision and recall, making it a reliable choice for realworld application.



6. Next Steps

- Explore deep learning models such as LSTMs, Bi-LSTMs, or Transformers (BERT, RoBERTa)
- Experiment with larger datasets to improve generalization
- Apply techniques like hyperparameter tuning and feature engineering

- Deploy the model as a real-time web application or API for practical usage

7. Possible Improvements

While the results are promising, there are areas for improvement:

- Better text preprocessing (handling sarcasm, slang, and multi-language data)
- Using contextual embeddings (BERT, GPT-based) instead of static embeddings
- Cross-validation for more robust evaluation
- Optimizing model training time for large-scale data

8. Conclusion

The project successfully demonstrated the potential of combining semantic embeddings with ensemble learning to address the problem of fake news detection. Random Forest achieved nearly 80% accuracy with strong F1-scores, outperforming Logistic Regression and Decision Trees. This indicates that feature-rich representations like Word2Vec, when paired with robust algorithms, can deliver reliable results. Future work should focus on leveraging deep contextual models and deploying the system for real-world use cases.